

Sequence Concatenation

Example 1 : Simple Concatenation (ASCII)

The caret operator after a sequence means concatenation:

$\langle a \rangle \frown \langle b \rangle$
 $\langle 1, 2 \rangle \frown \langle 3, 4 \rangle$
 $\langle x \rangle \frown \langle \rangle$

Example 2 : Concatenation with Variables

Concatenation with sequence variables:

$\langle x \rangle \frown s$
 $s \frown t$
 $\langle \rangle \frown t$

Example 3 : Cons Pattern

The cons pattern builds sequences incrementally:

$\langle x \rangle \frown \langle \rangle$
 $\langle x \rangle \frown \langle y \rangle \frown \langle z \rangle$
 $\langle 1 \rangle \frown \langle 2 \rangle \frown \langle 3 \rangle \frown \langle \rangle$

Example 4 : Nested Concatenation

Concatenation is associative:

$(\langle a \rangle \frown \langle b \rangle) \frown \langle c \rangle$
 $\langle a \rangle \frown (\langle b \rangle \frown \langle c \rangle)$
 $(s \frown t) \frown u$